Binary cubic V4 covers

Two exact eligibility gates, five canonical quartics, and evaluated subgroup maps

Status. The constructions and maps are verified for five listed curves. This report establishes a low-genus transfer route and a conditional asymptotic attack bound. A complete attack-cost comparison has not yet been measured.

1. Family theorem and bound

Let q=2^m, k=F_q, K=F_(q^3), and let E_(a,b)/K be the ordinary normalized curve Y^2+XY=X^3+aX^2+b with b nonzero. Let d be the inverse of 2(q+1) modulo q^3-1. The triangle-cover construction has a quotient isomorphic to E_(a,b) exactly when

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Tr_(K/F_2)(a) = Tr_(K/F_2)(b^d).
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The exponent exists because gcd(2(q+1),q^3-1)=1. For each b in K*, exactly one of the two normalized twists passes the trace test. Therefore the eligible count is **q^3-1 of 2(q^3-1)** ordinary normalized K-isomorphism classes. At q=16 it is 4095 of 8190. This count is exact for this construction; it is neither a global classification of cover routes nor a count of curves with a measured attack advantage.

A 2012 seminar abstract reports an approximately three-quarter density for the wider binary cubic (2,...,2) weak-cover class when the base field has even degree over F2. The present exact one-half count is a construction-specific lower bound, not the full weak-cover density.

Choose nonzero delta in ker Tr_(K/k), and t0+t0^q=delta. With t1=t0^q, t2=t0^(q^2), lambda=b^d, and A=delta^(q^2-q+1)lambda, put f0=A/(x+t0)+A^q/(x+t1) and f1=A^q/(x+t1)+A^(q^2)/(x+t2). The smooth normalization of y0^2+y0=f0, y1^2+y1=f1 is a connected V4 cover. Its three distinct simple branch poles give genus 3 by Riemann-Hurwitz. Frobenius cycles its three elliptic quotients, so the cover descends to k. The differing inertia labels at the three branch poles rule out a hyperelliptic involution; its canonical model is a smooth plane quartic.

2. Exact q=16 source curves

K=F_2[w]/(w^12+w^7+w^6+w^5+w^3+w+1). Integer coordinates and quartic coefficients are bitmasks in the basis 1,w,...,w^11. In every row Q=[7]G.

#E(K)	(a,b)	r	G	Q	LMFDB class
3972	(0,51)	331	(1226, 1765)	(1143, 3210)	3.16.a_a_aev
4006	(32,198)	2003	(199, 2783)	(738, 3018)	3.16.a_a_adn
4198	(32,116)	2099	(117, 3864)	(1053, 3821)	3.16.a_a_dx
4222	(32,853)	2111	(852, 2301)	(112, 1443)	3.16.a_a_ev

The 4006 and 4198 classes already had a positive Jacobian entry in LMFDB. The 3972 and 4222 classes were marked unknown on 27 October 2026; these quartics provide explicit positive witnesses for those classes. Equivalent equations elsewhere have not been excluded.

3. Explicit target quartics

For each row, F_N(X,Y,Z) is the dot product of its coefficient vector with the ordered monomials below. Each curve F_N=0 is smooth over F16, geometrically nonhyperelliptic, and has the predicted L-polynomial. Magma 2.30-1 receipts and canonical-map polynomials are retained beside the source files.

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(X^4, X^2Y^2, XY^3, Y^4, X^2YZ, XY^2Z, Y^3Z, X^2Z^2, XYZ^2, Y^2Z^2, XZ^3, YZ^3, Z^4)
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#E(K)	Coefficient vector c_0,...,c_12
3972	[1, 1641, 2890, 3362, 2623, 2890, 372, 372, 3362, 3159, 3363, 3158, 373]
4006	[1, 1641, 2890, 3362, 2623, 2890, 1821, 372, 3362, 2622, 3363, 1640, 373]
4198	[1, 1641, 2890, 1641, 2623, 2890, 2891, 372, 3362, 1640, 3363, 2890, 373]
4222	[1, 1641, 2890, 3159, 2623, 2890, 2890, 372, 3362, 1, 3363, 2622, 373]

The exact base-field affine model has the form L(u)D(x)=N(x), with L(U)=U(U+p)(U+v)(U+p+v). For the 4222 row:

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(u^4+2622u^2+3158u)(x^6+3362x^4+2623x^2+2623)
=3159x^5+3158x^4+2891x^3+1641x^2+1821x.
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All coefficients in this equation are F16 elements under the bitmask convention. The other exact affine equations are in descent_*.json, and audit.py independently checks their rational identities at every unramified x in K.

4. Subgroup-preserving transfer

Let pi_i:C_K -> E_i be the three degree-two quotient maps. For P in E_0(K), define Phi(P)=sum_i pi_i^*([P^(q^i)]-[O_i]). This is k-rational. Character orthogonality gives pi_0*Phi(P)=[2]P, so Phi is injective on every odd-order subgroup. The evaluator constructs degree-two quotient fibers, forms the divisor-class differences in the actual Sage quartic Jacobian, and sums the three Frobenius conjugates.

#E(K)	r	base n	Jacobian setup	map G,Q	checks
3972	331	17	14.68 s	5.06 s	all pass
4006	2003	36	14.12 s	5.56 s	all pass
4198	2099	7	14.76 s	5.95 s	all pass
4222	2111	10	14.93 s	5.65 s	all pass

Every row checks Phi(G) nonzero, [r]Phi(G)=0, Phi(Q)=7Phi(G), and [r]Phi(Q)=0. The displayed times are observed map stages on a shared host; they exclude a discrete-logarithm solve and cannot be compared directly with rho.

5. Attack bound and claim scope

Diem's theorem gives expected soft-O(q) for discrete logarithms in a nonhyperelliptic genus-three Jacobian over F_q. Combining that theorem with this explicit transfer gives a conditional asymptotic route. If an eligible source sequence has an odd prime subgroup r=Theta(q^3), generic rho uses Theta(q^(3/2)) group operations. The trace count does not prove that such large prime subgroups occur infinitely often. The Hasse bound controls total source order only: q^3+1-2q^(3/2) <= #E(K) <= q^3+1+2q^(3/2).

Magma's built-in plane-curve IndexCalculus rejects these q=16 quartics because its line-relation implementation requires q >= 4! = 24. Complete relation collection, linear algebra, target recovery, total cost, and matched source-subgroup rho remain unmeasured. Thus these four rows are transfer-eligible candidates, not established concrete speedups.

6. Prior work and novelty labels

The three labels must stay separate: **exact equation**, **eligible family**, and **attack method**. The 3972 and 4222 equations may be new catalog additions. The 4006 and 4198 classes are rediscoveries at isogeny-class level. The V4 cover type and genus-three transfer method have prior art. A SCIS 2024 abstract reports classification of all ordinary binary elliptic sources with extension degree at most nine outside the isogeny condition; a 2025 poster covers a domain that includes the normalized q=16 cubic source here. Exact equation-level overlap remains unresolved without their case tables. No global new-family or new-vulnerability claim follows from this audit.

Nart and Ritzenthaler (2009): genus-three characteristic-two V4 covers and elliptic quotients.

Murai, Shimura, Iijima, and Chao (2022): even-characteristic classification under the isogeny condition.

Iijima, Momose, and Chao (2013): references the earlier SCIS 2010/2011 cubic weak-cover papers.

Shimura seminar (2012): reports about three-quarter weak-cover density for even base-field degree.

SCIS program (2024): reports non-isogeny ordinary binary classification through degree nine.

Uesato poster (2025): gives a wider even-characteristic classification domain.

Diem (2012): expected soft-O(q) Jacobian DLP for nonhyperelliptic genus three.

Tian (2020): a separate cubic-extension genus-three transfer via isogenies.

7. Reproduction and formalization

The full derivation, all exact coefficients, replay scripts, Magma inputs and responses, Lean file, and candidate-specific prior-work record live in research/ghs_twisted_v4_triangle_20261027/ in merged PR #2735. TriangleGate.lean kernel-checks the q=16 exponent/count arithmetic and trivial doubling kernel for the four named odd prime orders. It does not formalize the function-field geometry or the quartic Jacobian computations.

8. Hess's known two-common-pole construction

Let V=ker Tr_(K/k), a two-dimensional k-space. Choose A,B in V* with A B=sqrt(b), and put f_i=A^(q^i)x+B^(q^i)/x for i=0,1,2. The three f_i sum to zero. The connected V4 cover y0^2+y0=f0, y1^2+y1=f1 descends to k; its two common branch poles have full V4 inertia and total different 12, so its smooth normalization has genus three. Its first elliptic quotient is the normalized E_(a,b) precisely for the trace-zero twist Tr_(K/F2)(a)=0.

This is Hess's published double-trace-zero branch. The exact all-q product count was already proved in this repository's earlier Hess audit. The calculation below restates it to make the overlap with the triangle gate precise.

The product test is a binary-quadratic isotropy test: Q_s(A)=Tr_(K/k)(s A^q A^(q^2))=0 for s=sqrt(b). Over an algebraic closure, the three conjugates of A have coordinates (u,v,u+v); the map from s to the three coefficients of Q_s is invertible. Counting double and distinct rational linear factors gives the exact all-q product count below.

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#(V*V*) = (q^2-1)(q+2)/2.
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The all-q overlap has a closed form. Put e=(q+1)^(-1) modulo q^3-1 and t=s^e. Since b=s^2, the triangle trace is the absolute trace of t. Scaling s by c in k* scales t by sqrt(c). The projective product classes with relative trace t=0 are exactly t^(q+1) for t in V*, giving q+1 classes; the remaining q(q+1)/2 classes each contribute q/2 trace-one scalars. Hence exactly q^2(q+1)/4 source models pass Hess while failing the triangle gate.

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Union of the two gates: q^3-1+q^2(q+1)/4 of 2(q^3-1).
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At q=16, the Hess gate has 2295 b values, the new overlap count adds 1088 models, and the union is **5183 of 8190**. This is an exact cover-eligibility count and a lower bound for the wider cover universe; it is not an attack-cost count or a global vulnerability classification. The overlap formula is new to this audit; its publication novelty is unresolved.

9. Exact two-pole witness and subgroup map

In the pinned F_(2^12) field, A=1079, B=2852, a=0, b=3399, and E: Y^2+XY=X^3+b. It has #E(K)=4132=4·1033, G=(2305,2550), and Q=(2273,2935)=7G. Its descended affine function field is

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x^2(u^4+2622u^2+3158u)=1821x^4+373x^3+1640x+1.
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Magma 2.30-1 confirms genus three, a smooth nonhyperelliptic canonical quartic, and L-polynomial 1+35T^3+4096T^6. Its canonical quartic has the coefficient vector below in the monomial order X^4,X^3Y,X^2Y^2,Y^4,X^2YZ,XY^2Z,X^2Z^2,XYZ^2,YZ^3,Z^4.

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[1, 3159, 1821, 1640, 3363, 1821, 3362, 3159, 1820, 3362]
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Sage evaluates the quotient pullback and Frobenius trace in this quartic's actual Jacobian: Phi(G) is nonzero, [1033]Phi(G)=0, and Phi(Q)=7Phi(G). Jacobian setup was 9.21 s and the two map evaluations were 5.09 s. A full index-calculus and matched-rho cost is still unmeasured.

LMFDB class 3.16.a_a_bj already records a positive Jacobian. Hess (2004) covers the factor branch. This is a new q=16 exact map witness in this audit, while its family and isogeny class are known. Its exact equation has not been checked against every earlier case table.

TwoPoleGate.lean kernel-checks the q=16 count arithmetic and the doubling kernel on the 1033-order subgroup; it does not formalize the field-product proof or geometric map.